BRAINSTORM GROUP (BSG)

MAT101 E-test Question

1. $Given\ f(x) = x + 3, g(x) = x^2 + 1\ and\ h(x) = x.\ Find\ foh$ A. $x^2 + 1$ B. $x^2 + 4$ C. $x + 1$ D. $x^2 + 2$
2. $Given\ f(x) = x + 3, g(x) = x^2 + 1\ and\ h(x) = x.\ Find\ goh$ A. $x^2 + 1$ B. $x^2 + 4$ C. $x + 1$ D. $x^2 + 2$
3. $Given\ f(x) = x + 3, g(x) = x^2 + 1\ and\ h(x) = x. Find\ gof$
4. $Given\ f(x) = x + 3, g(x) = x^2 + 1\ and\ h(x) = x. Find\ hof$
5. $Given\ f(x) = x + 3, g(x) = x^2 + 1\ and\ h(x) = x. Find\ hog$
6. $Given\ f(x) = x + 3, g(x) = x^2 + 1\ and\ h(x) = x.\ Find\ fog$ A. $2x^3 + 4$ B. $2x^2 + 4$ C. $x^2 + 4$ D. $2x^2$
7. $Given\ f(x) = x + 3, g(x) = x^2 + 1\ and\ h(x) = x.\ Find\ fogoh$
8. $Evaluate\ |x - 3| > 1$ A. $x > 2$ B. $x < 2$ C. $x > 4$ D. $A\ \&\ B$
9. $Evaluate\ |2x - 5| < 3$ A. $x < 4$ B. $1 < x < -4$ C. $x > 1$ D. $A\ \&\ B$
10. $Evaluate\ |2 - 4x| \geq 6$ A. $x \geq 2$ B. $2 \leq x > -1$ C. $x \leq -1$ D. $A\ \&\ B$
11. $Express\ cos^2\theta\ in\ terms\ of\ multiole\ angles$ A. $2[cos2\theta - 1]$ B. $2^{-1}[cos2\theta - 1]$ C. $2[cos2\theta + 1]$ D. $2^{-1}[cos2\theta + 1]$
12. $Find\ f(x) = \sqrt{x+4}\ when\ f(x) = \{0\}$ A. -2 B. 0 C. 2 D. $\sqrt{2}$
13. $Find\ f(x) = \sqrt{x+4}\ when\ f(x) = \{-1\}$
14. $Find\ f(x) = \sqrt{x+4}\ when\ f(x) = \{-2\}$ A. $\sqrt{-1}$ B. 0 C. 2 D. $\sqrt{2}$
15. $Find\ f(x) = \sqrt{x+4}\ when\ f(x) = \{-3\}$ A. i B. -1 C. 1^2 D. i^2
16. $Find\ f(x) = \sqrt{x+4}\ when\ f(x) = \{-4\}$
17. $Given\ that\ f(x) = \sqrt{x+4}. Compute\ f(-2)$ A. 1 B. 0 C. $\frac{1}{\sqrt{2}}$ D. i
18. $Given\ that\ f(x) = \sqrt{x+4}. Compute\ f(-4)$ A. $i + 2$ B. 0 C. 4 D. i
19. $What\ is\ the\ value\ of\ a\ given\ a + ib = \frac{1-i}{2+i}$

    A. $-\frac{1}{5}$ B. $\frac{3}{5}$ C. $-\frac{3}{5}$ D. $\frac{1}{5}$
20. Given that $Z_1 = 3 + 2i$ and $Z_2 = 4 - 5i$, evaluate $Z_1 Z_2$

    A. $2 - 7i$ B. $22 - 7i$ C. $2 + 7i$ D. $22 + 7i$
21. $If\ Z_1 = 1 - 3i\ and\ Z_2 = -2 + 5i, determine\ \frac{Z_1 Z_2}{Z_1 + Z_2}\ in\ the\ form\ a + ib$

    A. $-\frac{9}{5} + \frac{37}{5}i$ B. $\frac{9}{5} - \frac{37}{5}i$ C. $-\frac{12}{25} - \frac{13}{25}i$ D. $\frac{9}{25} - \frac{37}{25}i$
22. Given the universal set $M = \{1, 2, 3, 4, 5, 6, 7, 8\}$ where $A = \{1, 3, 5, 7\}. B = \{1, 2, 5, 7\}\ and C = \{3, 6, 7, 8\}$ list the members of $A^c \cap B \cap C^c$ A. {1} B. {2} C. {3} D. {4}
23. If $X \cap Y = \emptyset$ then X and Y are said to be A. finite sets B. disjoint sets C. Null sets
24. $Solve\ x^2 + 8x + 16 > 0$ A. $x > -4$ B. $x < 4$ C. $x \geq 2$ D. $x \geq -4$
25. If set $F = \{x : (2 < x \leq 5) \cup (9 < x < 12)\}$. List the members of set F. A. {3, 4, 5, 10, 11} B. {3, 4, 5, 7, 11} C. {2, 3, 4, 5, 10}
26. Determine the modulus, argument of $2 + 3i$
27. Determine the modulus, argument of $2 + 4i$.

    A. $2\sqrt{5},\ 63.43°$ B. $2\sqrt{10},\ 63.43°$ C. $2\sqrt{5},\ 26.57°$ D. $2\sqrt{10},\ 26.57°$
28. Solve the inequality $2x - 3 < x + 7$ A. $x - 10 < 0$ B. $x < 10$ C. $x > -10$ D. $x - 10 > 0$
29. Determine the value of x satisfying $|x + 3| \geq 7$ A. $x \leq -4, x \geq 10$ B. $x \leq -10, x \geq 4$ C. $x \leq -10, x \geq -4$ D> $x \leq -4, x \geq 10$

30. $Simplify\ \sqrt{20}+\sqrt{45}+\sqrt{125}-2\sqrt{80}$ A. $2\sqrt{5}$ B. $3\sqrt{5}$ C. $4\sqrt{5}$ D. $5\sqrt{5}$
31. $Simpilfy\ 3\sqrt{2}-\sqrt{32}+\sqrt{50}+\sqrt{98}$ A. $5\sqrt{2}$ B. $6\sqrt{2}$ C. $10\sqrt{2}$ D. $11\sqrt{2}$
32. $Find\ \frac{Z_1}{Z_2}\ if\ Z_1=2-4i\ and\ Z_2=6+7i$

A. $\left(\frac{16}{85}+i\frac{38}{85}\right)$ B. $\left(\frac{15}{85}-i\frac{38}{85}\right)$ C. $-\left(\frac{16}{85}-i\frac{38}{85}\right)$ D. $-\left(\frac{16}{85}+i\frac{38}{85}\right)$

33. $If\ Z_1=1-3i\ and\ \ Z_2=-2+5i, determine\ \frac{Z_1Z_2}{Z_1+Z_2}\ in\ the\ form\ a+ib$

A. $-\frac{9}{5}+\frac{37}{5}i$ B. $\frac{9}{5}-\frac{37}{5}i$ C. $-\frac{12}{25}-\frac{13}{25}i$ D. $\frac{9}{25}-\frac{37}{25}i$

34. Given that $Z_1=2+3j$ and $Z_2=3-4j$, evaluate Z_2-Z_1

A. $1+7j$ B. $1-7j$ C. $-1+7j$ D. $-1-7j$

35. Given $Z_1=1+2i,\ Z_2=4-3i,\ Z_3=-2+3i$, evaluate $|Z_1+Z_2|-|Z_3|\times\sqrt{2}$

A. $\sqrt{10}$ B. 10 C. $2\sqrt{5}$ D. 0

36. If $Z_1=1+2i\ and\ Z_2=4-3i$, what is the value of $|Z_2-Z_1|$?

A. $\sqrt{34}$ B. $\sqrt{16}$ C. $2\sqrt{5}$ D. 4

37. Determine the modulus, argument of $-5-2i$.

A. $5.385, -158.2°$ B. $5.385,\ 26.57°$ C. $4.583, -158.2°$ D. $4.583,\ 26.57°$

38. $Given\ Z_1=(3+2i) and\ Z_2=(4-5i), evaluate\ Z_1Z_2$
39. _____ numbers ranges from $-\infty$ to $+\infty$

A. real B. integer C. complex D. natural

40.

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